

CTEs (Common Table Expressions) – Short Notes

What is a CTE?

A **CTE (Common Table Expression)** is a **temporary named result set** that can be used within a SQL query. It makes complex queries easier to read and maintain.

A CTE exists **only for the duration of the query**.

Syntax

```
WITH cte_name AS (  
    SELECT column1, column2  
    FROM table_name  
)  
SELECT *  
FROM cte_name;
```

Example

Suppose we have an `employees` table.

```
WITH high_salary AS (  
    SELECT *  
    FROM employees  
    WHERE salary > 50000  
)  
SELECT *  
FROM high_salary;
```

This creates a temporary table named `high_salary` and then selects from it.

Advantages of CTEs

- Makes queries easier to read
 - Avoids repeating the same subquery
 - Simplifies complex SQL queries
 - Can improve query organization
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Subqueries – Short Notes

What is a Subquery?

A **subquery** is a **query written inside another SQL query**.

The inner query executes first, and its result is used by the outer query.

Syntax

```
SELECT column_name
FROM table_name
WHERE column_name operator (
    SELECT column_name
    FROM another_table
);
```

Example 1: Subquery in WHERE

Find employees earning more than the average salary.

```
SELECT name, salary
FROM employees
WHERE salary >
(
    SELECT AVG(salary)
    FROM employees
);
```

Example 2: Subquery in FROM

```
SELECT AVG(salary)
FROM
(
    SELECT salary
    FROM employees
) AS emp;
```

Example 3: Subquery in SELECT

```
SELECT
    name,
    salary,
    (SELECT AVG(salary) FROM employees) AS average_salary
FROM employees;
```

Types of Subqueries

1. Single-row Subquery

Returns only one row.

Example:

```
SELECT *
FROM employees
WHERE salary >
(
    SELECT AVG(salary)
    FROM employees
);
```

2. Multiple-row Subquery

Returns multiple rows.

Example:

```
SELECT *
FROM employees
WHERE department_id IN
(
    SELECT department_id
    FROM departments
);
```

3. Correlated Subquery

The inner query depends on the outer query and executes once for each row.

Example:

```
SELECT e1.name, e1.salary
FROM employees e1
WHERE salary >
(
    SELECT AVG(e2.salary)
    FROM employees e2
    WHERE e1.department_id = e2.department_id
);
```

CTE vs Subquery

CTE	Subquery
Created using <code>WITH</code>	Written inside another query
Improves readability	Can become difficult to read if nested
Can be referenced multiple times in the same query	Usually used once
Better for complex queries	Good for simple queries

Summary

Term	Description
CTE	A temporary named result set created using WITH
Subquery	A query inside another query
Single-row Subquery	Returns one row
Multiple-row Subquery	Returns multiple rows
Correlated Subquery	Depends on the outer query and runs for each row

Remember

- **CTE = Temporary named table used within a query.**
- **Subquery = Query inside another query.**
- **CTEs improve readability and can be reused in the same query.**
- **Subqueries are commonly used in **WHERE** , **FROM** , and **SELECT** clauses.**